



EENG 385 - Electronic Devices and Circuits
BJT Curve Tracer: Schmitt Trigger Relaxation Oscillator
Lab Document

Outcome and Objectives

The outcome of this lab is to analyze, simulate and assemble a circuit that generates a periodic square wave using a circuit with positive feedback and compare the expected behavior of the circuit from each mode of analysis. Through this process you will achieve the following learning objectives:

- Analyze and design a circuit containing resistors and op amps.
- Use a software tool to perform time and frequency domain analysis of an electronic circuit.
- Assemble a circuit on a PCB using the equipment in the laboratory.
- Use laboratory test and measurement equipment to analyze electronic circuits.

Analysis: Schmitt Trigger Relaxation Oscillator

During today's lab you will build the circuit shown in Figure 1, a Schmitt Trigger Relaxation Oscillator. At the outset, I need to be clear, the engineering objectives of this lab could be accomplished with a 555 Timer. However, the education objective of this lab is to explore the use of positive feedback in an op amp to generator oscillatory behavior.

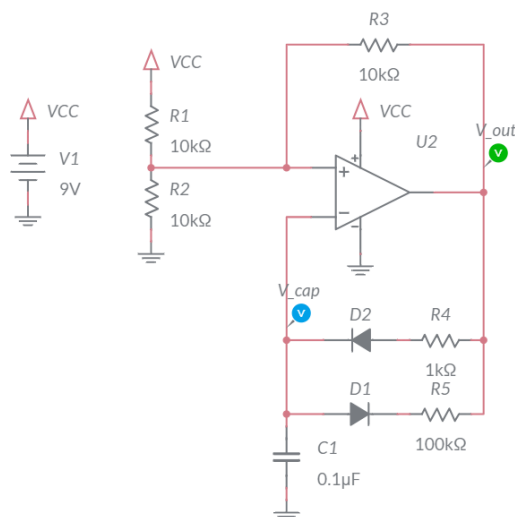


Figure 1: The Schmitt Trigger Relaxation Oscillator. The output is taken from the op amp output.

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In order to analyze the circuit in Figure 1, you need to understand the behavior of a Schmitt Trigger – a circuit with one input, V_{in} , and one output, V_{out} . The Schmitt Trigger compares V_{in} against two voltage levels, V_{high} and V_{low} and assigns the output as follows:

- If ($V_{in} > V_{high}$), then $V_{out} = VCC$
- If ($V_{in} < V_{low}$), then $V_{out} = 0V$
- If ($V_{low} < V_{in} < V_{high}$), then V_{out} remains unchanged

The behavior is captured in Figure 2, a graph of V_{in} vs. V_{out} .

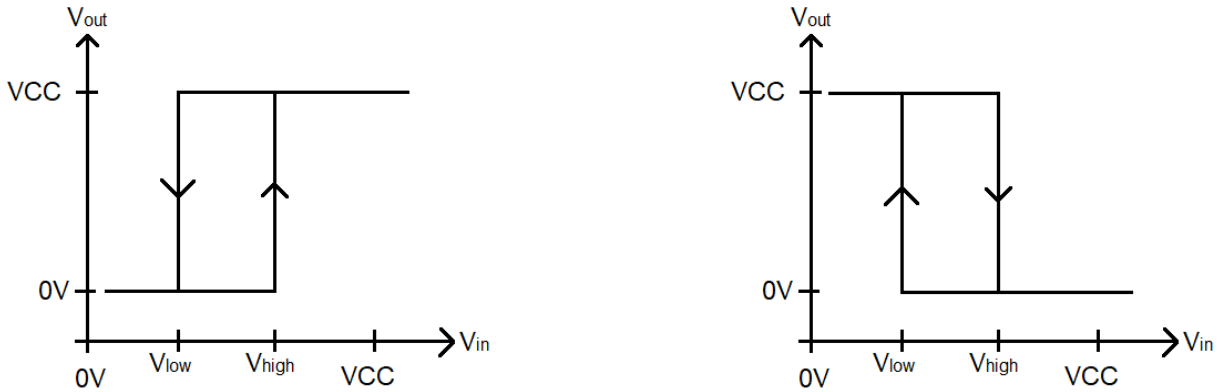


Figure 2: Input, output relationship for a Schmitt Trigger is characteristic of a system with hysteresis. Left a normal Schmitt Trigger and right, an inverting Schmitt trigger.

Let's focus on the curve to the left in Figure 2. To understand this figure, use your finger to point to the region of the horizontal axis (the V_{in} axis) which corresponds to $V_{in} > V_{high}$. Look above this region and you will see a horizontal line corresponding to $V_{out} = VCC$. Likewise for $V_{in} < V_{low}$ you should see that $V_{out} = 0V$. The tricky part of the graph is when V_{in} is between V_{low} and V_{high} . In this region, V_{out} holds on to the value it had before V_{in} entered the region between V_{low} and V_{high} .

For example, let's say $V_{in} = VCC$, so we know for sure that $V_{out} = VCC$. Now, let V_{in} decrease slowly, all the while V_{out} will stay at VCC . At some point V_{in} will become equal to and then just a smidge lower than V_{high} . At this point, V_{high} will remain unchanged at stay at VCC . As V_{in} decreases towards V_{low} , V_{out} will remain at VCC . Only when V_{in} decreases, just a smidge, below V_{low} will V_{out} switch its value to $0V$. This transition is represented in Figure 2 by the downward arrow on the line connecting VCC to $0V$ at $V_{in} = V_{low}$.

Similarly, as V_{in} increases from $0V$ towards V_{high} , the output, V_{out} , will stay at $0V$. Only when V_{in} increases above V_{high} will the output change to VCC . This transition is represented in Figure 2 by the upward arrow on the line connecting $0V$ to VCC at $V_{in} = V_{high}$.

Systems with hysteresis are well suited to reject noise from signals. Consider the example shown in Figure 3 where the top analog signal (red line) is converted into digital signal. The middle signal shows how the analog is converted when a single threshold (the grey line between V_{high} and V_{low}) is used to classify the analog signal as 1 when it is above the grey line and 0 when the analog signal is below the grey line. Any noise on the analog signal will cause this scheme to make several digital transitions.

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The bottom graph in Figure 3 shows how the analog signal is converted when two thresholds, V_{high} and V_{low} , and hysteresis are used to classify the analog signal. In this scheme, the signal is classified as 0 when the analog signal is below V_{low} and classified as 1 when the analog signal is above V_{high} . When the analog signal is between the two thresholds, it retains its previous value. The resulting digital signal is much more immune to noise on the analog signal.

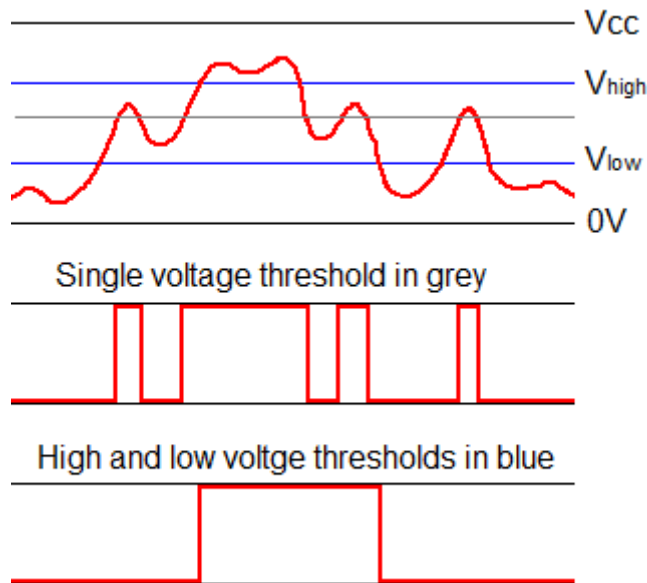


Figure 3: The top signal (in red) is converted into a digital signal using the single grey threshold (middle) and the 2 blue thresholds (lower).

Figure 4 shows a Schmitt Trigger built using an op amp and resistors. Can you identify the parts of this circuit used in the circuit shown in Figure 1? Note, the resistors are the same in both figures.

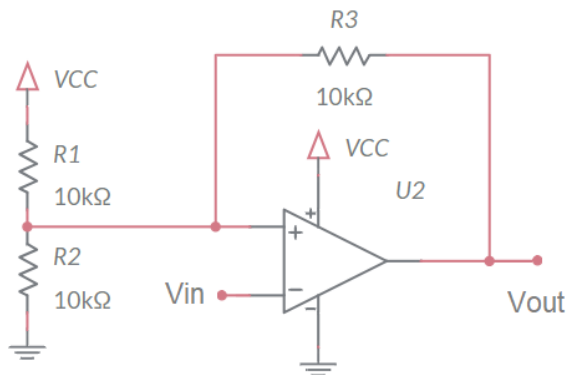


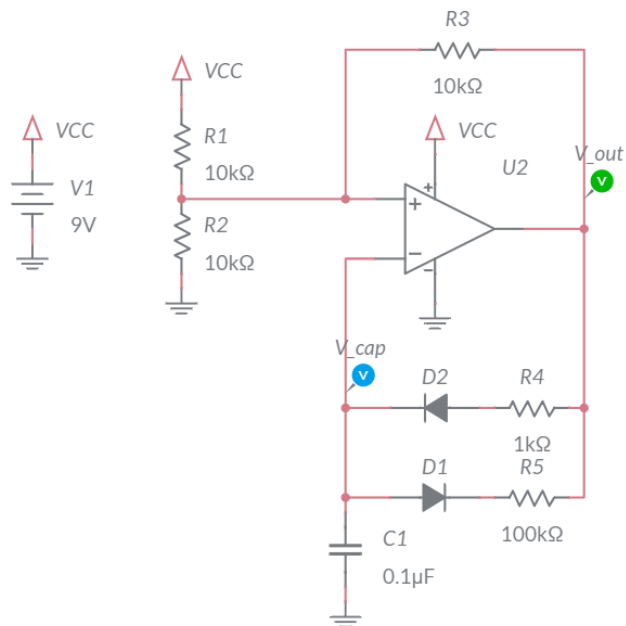
Figure 4: A Schmitt Trigger built using an op amp. Don't forget to add a 9V power supply!

Answer the following questions to understand the behavior of the circuit in Figure 4.

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1. Assume $V_{out} = 9V$. What is the voltage at the non-inverting input of the op amp? Call this voltage V_{high} . (Hint: Combine the parallel resistors then use Ohm's law.)
2. Assume $V_{out} = 0V$. What is the voltage at the non-inverting input of the op amp? Call this voltage V_{low} . (Hint: Combine the parallel resistors then use Ohm's law.)
3. Imagine V_{in} is at $0V$. What is V_{out} ? (Hint: The voltage on the non-inverting input of the op amp is greater than V_{low} .)
4. Imagine increasing V_{in} from $0V$ to just below V_{high} . What is V_{out} ? (Hint: Use V_{out} from the Q3, since you are starting at $0V$, to determine the value of non-inverting op amp input from Q1.)
5. Imagine V_{in} continues to increase and goes just above V_{high} . What is V_{out} ?
6. Imagine V_{in} is at $9V$. What is V_{out} ? (Hint: The voltage on the non-inverting input of the op amp is at least V_{high} .)
7. Imagine decreasing V_{in} from $9V$ to just above V_{low} . What is V_{out} ? (Hint: Use V_{out} from the Q6, since you are starting at $9V$, to determine the value of non-inverting op amp input from Q2.)
8. Imagine V_{in} continues to decrease and goes just below V_{low} . What is V_{out} ?
9. Use this information to draw V_{in} vs. V_{out} hysteresis diagram similar to Figure 2. Label the V_{in} -axis with the voltage values for V_{low} and V_{out} found in Q1 and Q2.

Now with your solid understanding of the Schmitt Trigger circuit in Figure 4, it is time to apply your understanding to the operation of the larger circuit in Figure 1. Do this by working through the following questions.



10. Assume $V_{out} = 9V$. In this case, there is a path for electrical flow from V_{out} through R4, through the (forward biased) diode D2, to the capacitor C1 to ground. Diode D1 is reverse biased so it eliminates the $100k\Omega$ resistor from the circuit. What is the time constant for this charging? Hint: You can replace the forward biased diode D2 with a wire.
11. Use this time constant to write an equation describing the voltage on capacitor C1.
12. Compute the time required for capacitor C1 to charge from V_{low} to V_{high}

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- Set the equation derived in Q11 equal to V_{high} and solve for t . Represent your answer in microseconds and round to the nearest integer.
 - Set the equation you derived in Q11 equal to V_{low} and solve for t . Represent your answer in microseconds and round to the nearest integer.
 - To derive the time to charge C1 from V_{low} to V_{high} , subtract the time to get to V_{high} from the time to get to V_{low} .
13. Assume $V_{out} = 0V$. In this case, there is a path for electrical flow from the charged plate of the capacitor C1 through the (forward biased) diode D1, through resistor R5 to $V_{out} = 0V$. Diode D2 is reverse biased so eliminates the $1k\Omega$ resistor from the circuit. What is the time constant for this discharging? (*Hint: You can replace the forward biased diode D1 with a wire.*)
14. Use this time constant to write an equation describing the voltage on capacitor C1.
15. Compute the time required for the capacitor C1 to discharge from V_{high} to V_{low} as follows.
- Set the $V_c(t)$ equation equal to V_{high} and solve for t . This equation gives the time to discharge from 9V to V_{high} . Represent your answer in milliseconds and round to three significant figures.
 - Set the $V_c(t)$ equation equal to V_{low} and solve for t . This equation gives the time to discharge from 9V to V_{low} . Represent your answer in milliseconds and round to three significant figures.
 - To derive the time to discharge C1 from V_{high} to V_{low} , subtract the time to get to V_{high} from the time to get to V_{low} .

Note, the capacitor C1 is connected to the inverting input of the op amp in Figure 1. Thus, the charge on this capacitor plays the role of V_{in} in Figure 4 and in the graph produced in Q9. Use your answers to these and other questions to answer the following two questions. Make sure to fill in the blanks and choose an option when a pair of **items** are separated by a “/”.

16. When $V_{out} = 9V$, then the capacitor C1 is **charging/discharging** from V_{low} to V_{high} . The capacitor takes ____ μs to go from V_{low} to V_{high} . When the charging/discharging capacitor's voltage exceeds V_{low}/V_{high} then $V_{out} = 0V$. When $V_{out} = 0V$, the capacitor will start to **charge/discharge**.
17. When $V_{out} = 0V$, then the capacitor C1 is **charging/discharging** from V_{high} to V_{low} . Now the capacitor takes ____ ms to go from V_{high} to V_{low} . When the charging/discharging capacitor's voltage drops below V_{low}/V_{high} , then $V_{out} = 9V$. When $V_{out} = 9V$, the capacitor will start to **charge/ discharge**.

Thus, V_{out} oscillates between 9V and 0V with the characteristics of the waveform determined by the time required to charge and discharge the capacitor C1. Use the information from this section to fill in the **Analysis** columns of Table 3 and Table 4.

Simulation: Schmitt Trigger Relaxation Oscillator

Build the circuit in Figure 1 using MultiSim Live. Make sure to attach probes to the output of the op amp and the inverting input of the op amp. The parts list is similar to last weeks; just in case, the bill of materials is given in Table 1.

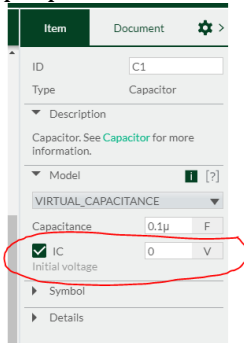
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Table 1: The parts list for the Schmitt Trigger Relaxation Oscillator.

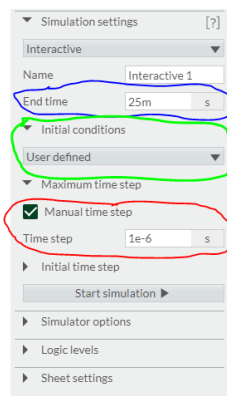
Component	Tool	Name
DC Voltage Supply	Sources	DC Voltage
Ground	Schematic connectors	Ground
VCC	Schematic connectors	Connector
Resistor	Passive	Resistor
Capacitor	Passive	Capacitor
Diode	Diodes	Diode
Op amp	Analog	5 Terminal Op amp

Once you have completed the schematic, use the export option in the main menu to output a PNG file of the schematic. Then, make some important changes in order for the simulation to run correctly.

- Set the initial voltage on the 0.1 μF capacitor to 0V. Do this by opening the 0.1 μF capacitor's properties, checking the box next to IC and entering the value "0" in the text box.



- Set the simulation time step to 1 μs . Do this by opening the document properties menu, expanding the **Maximum time step** option, checking the "Manual time step" checkbox and filling "1e-7" in the "Time step" text box.
- In this same menu, change the simulation End time to 25ms by filling "25m" in the "End time" text box.
- In this same menu, change the Initial Conditions to "User defined" by selecting this option from the pull-down menu.



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Include one wavelength of the output and capacitor voltage in your answers. You can use the export option in the main menu to output a PNG file.



Figure 5: Simulation output of the circuit shown in Figure 1.

Use the V_{out} waveform to fill in the **Simulation** column in Table 3. You may want to use the cursors function available in the Item tab. If you do not see this tab, double click on a blank area of the timing diagram to make it appear.

Use the V_{cap} waveform to fill in the **Simulation** column in Table 4. Fill in the V_{high} row with the highest voltage appearing on the V_{cap} waveform. Fill in the V_{low} row with the lowest voltage appearing on the V_{cap} waveform.

Empirical: Schmitt Trigger Relaxation Oscillator

This week, you will be soldering in the components associated with the SCHMITT TRIGGER RELAXATION OSCILLATOR subsystem in Figure 6. You should solder in all the components associated with this subsystem and the resistor R10 for the RESET CIRCUIT. This lab document contains cursory coverage of the assembly process. You can find much more detail in the Assembly Guide posted on Canvas.

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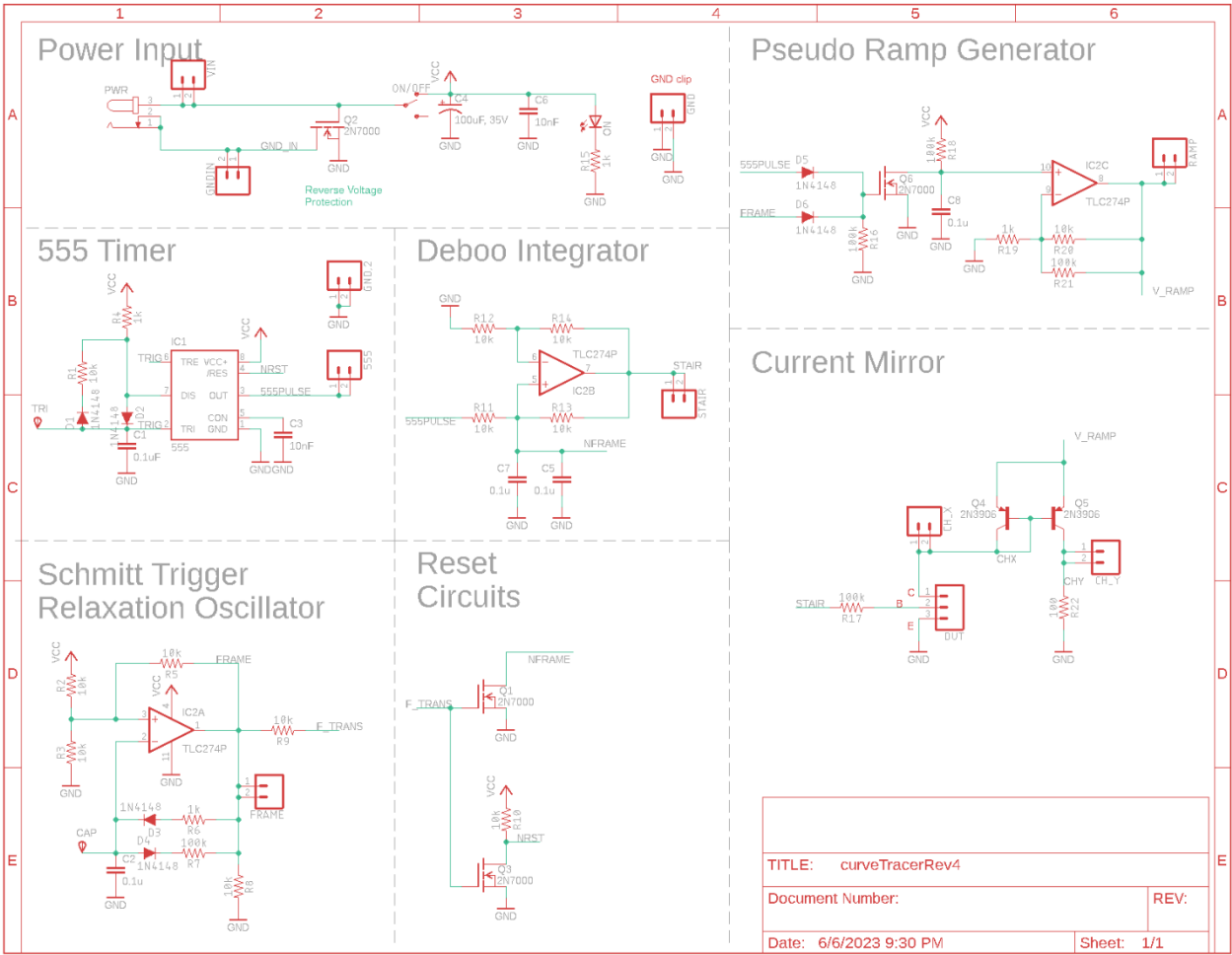
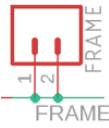
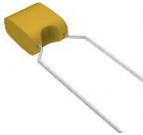


Figure 6: The schematic for the overall BJT curve tracer.

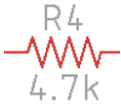
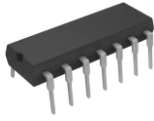
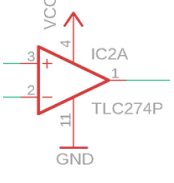
Part Identification

To make sure you can positively identify all the elements in the schematic, complete Table 2 by filling in the **Match** column with the letter corresponding to the **Schematic Symbol** for a **Physical Part**.

Table 2: Match the schematic symbol with the corresponding part.

Schematic Symbol	Match	Physical Part
 A		
 B		

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C 	
D 	
E 	A 

Testing

Once you have completed assembly of your SCHMITT TRIGGER RELAXATION OSCILLATOR subsystem, perform the following test.

- 1) Power up an oscilloscope. Attach a probe to Channel 1 and configure it as follows.

Ch 1 probe	FRAME test point (Vout)
Ch 1 ground clip	GND test point
Horizontal (scale)	1 ms
Ch 1 (scale)	1 V or 2 V (whatever fits better)
Ch 2 probe	Inverting input of op amp (Vcap)
Ch 2 (scale)	Same as Channel 1
Trigger mode	Auto
Trigger source	Ch1
Trigger slope	↑
Trigger level	4.5V

- 2) Set the GND reference of Ch 1 and Ch 2 to the lowest visible reticule – the waveforms will overlap the same as they did in the MultiSim simulation. Set the horizontal position of the trigger to the left most visible reticule. Note, the op amp output is sent to the FRAME test point. The capacitor charge is available by attaching an oscilloscope probe to the CAP test point shown in Figure 8.

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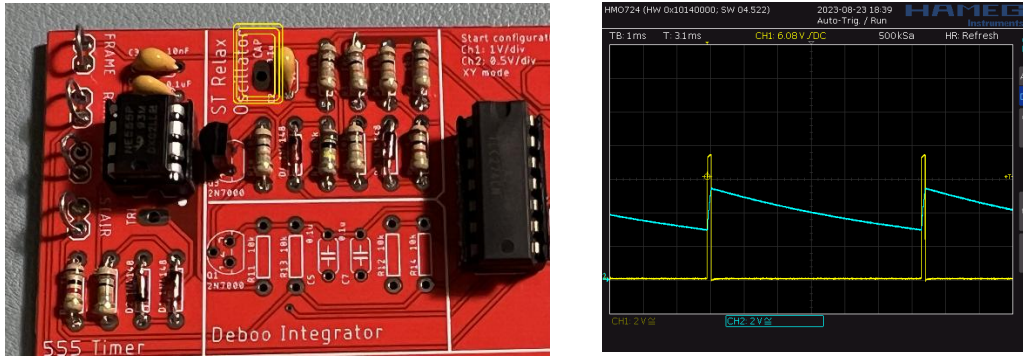


Figure 7: (Left) You can probe the capacitor voltage at the yellow circled test point. (Right) An oscilloscope trace showing the two output you need to capture. Note that this image was captured on a Rhode&Schwarz HMO724.

- 3) Take a screen shot of the of the V_{out} and V_{cap} waveforms and include them in your lab report. Screen shot the oscilloscope traces on USB. Cell phone pictures will lose points.
[Save] → Save → Format → 24-bit Bit... (*.bmp) [Save] → Save → Press to Save

You may want to apply the Acquire function to average 32 waveforms together to smooth the waveforms.

Use the data collected from the oscilloscope to fill out the **Assemble** columns in Table 3, and Table 4.

Comparison: Schmitt Trigger Relaxation Oscillator

Complete the **Analysis**, **Simulation** and **Empirical** columns of the following tables using the information you found throughout this lab. Represent your answer to 3 significant figures using the units given in parenthesis in the **Quantity** column. You will need this table in later labs, so keep it handy.

Table 3: Summary of V_{out} behavior in the Schmitt Trigger Relaxation Oscillator.

Quantity	Analysis	Simulation	Empirical
Time high (us)			
Time low (us)			
Period (us)			
Frequency (Hz)			
Duty Cycle			

Table 4: Summary of the $0.1\mu F$ capacitor voltage in the Schmitt Trigger Relaxation Oscillator.

Quantity	Analysis	Simulation	Empirical
V_{high} (volts)			
V_{low} (volts)			

Turn In: Schmitt Trigger Relaxation Oscillator

- 1) Make a record of your response to numbered items below and turn in a single lab report for your team on Canvas using the instructions posted there.
- 2) Include the names of both team members at the top of your solutions.
- 3) Use complete English sentences to introduce what each of the items listed below is and how it was derived.

Hint, use Ctrl+click to follow links. This also works for all the Figures and Tables in these labs.

Analysis: Schmitt Trigger Relaxation Oscillator

[Steps](#) 1 – 17.

Simulation: Schmitt Trigger Relaxation Oscillator

[Schematic](#) (use Export -> Schematic Image).

[Timing](#) diagram (use Export -> Grapher Image).

Empirical: Schmitt Trigger Relaxation Oscillator

Complete Table 2.

[Screen capture](#) of Schmitt Trigger Relaxation Oscillator output

Comparison: Schmitt Trigger Relaxation Oscillator

Table 3 and Table 4. Compare timer output in different models.